Constrained Convex Optimization

$$\min: f(x)$$
$$\text{s.t.: } x \in \mathcal{X}$$

GD: $x_{t+1} = x_t - \eta \nabla f(x_t)$

Issue: x_{t+1} may not be in $\mathcal{X}$.

Projected Gradient Descent

$$\min: \ f(x)$$
$$\text{s.t.} \quad x \in \mathcal{X}$$

<u>PGD</u>:

$$y_{t+1} = x_t - \eta \nabla f(x_t)$$
$$x_{t+1} = \mathrm{Proj}_{\mathcal{X}}(y_{t+1}) = \Pi_{\mathcal{X}}(y_{t+1}) = P_{\mathcal{X}}(y_{t+1})$$

Equiv: $x_{t+1} = \mathrm{Proj}_{\mathcal{X}}\left(x_t - \eta \nabla f(x_t)\right).$

<u>Oracle</u>

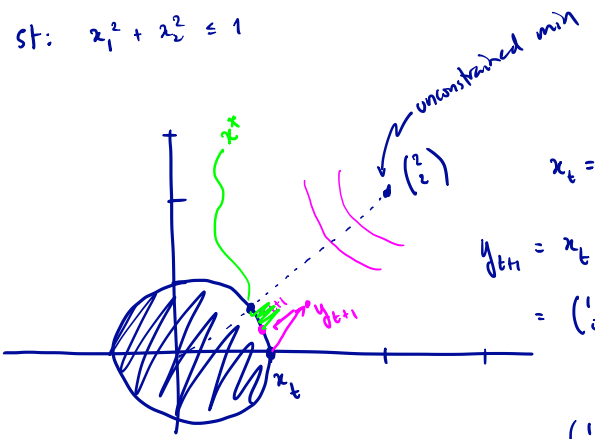
$$x \longrightarrow \square \longrightarrow \begin{matrix} \nabla f(x) \\ f(x) \end{matrix}$$

Now we need a <u>projection method</u>.

PGD — An Example

$$\min: \; x_1^2 - 4x_1 + x_2^2 - 4x_2$$

$$\text{st}: \; x_1^2 + x_2^2 \le 1$$



$$x_t = \begin{pmatrix}1\\0\end{pmatrix}$$

$$y_{t+1} = x_t - \eta \nabla f(x_t)$$

$$= \begin{pmatrix}1\\0\end{pmatrix} - \eta \begin{pmatrix}2x_1 - 4\\2x_2 - 4\end{pmatrix}\Bigg|_{\begin{pmatrix}1\\0\end{pmatrix}}$$

$$= \begin{pmatrix}1\\0\end{pmatrix} - \eta \begin{pmatrix}-2\\-4\end{pmatrix} = \begin{pmatrix}1 + 2\eta\\4\eta\end{pmatrix}$$

Take $\eta = 1/3$: $y_{t+1} = \begin{pmatrix}5/3\\4/3\end{pmatrix}$ $\quad x_{t+1} = \dfrac{\begin{pmatrix}5/3\\4/3\end{pmatrix}}{\left\|\begin{pmatrix}5/3\\4/3\end{pmatrix}\right\|_2}$

PGD — Convergence Rates

Convergence rate guarantees are the same as for unconstrained case.

f smooth : $O(1/T)$

f smooth & strongly convex : Linear $O\left((1-m/M)^T\right)$

Step size: same as for unconstrained : $\eta = 1/M$.